

BUSINESS REPORT

TIME SERIES FORECASTING

(SPARKLING)

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G4 PGDSBA.O.May 23.A

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Problem:

The data of different types of wine sales in the 20th century is to be analyzed. Both of these data are from the same company but of different wines. As an analyst in the ABC Estate Wines, you are tasked to analyse and forecast Wine Sales in the 20th century.

Data set for the Problem: Sparkling.csv

1. Read the data as an appropriate Time Series Data and plot the Data.

All the necessary packages for analysis of Time Series have been downloaded and data of Sparkling Wine Sales having 187 rows ranging from 01.01.1980 to 01.07.1995 has been read as Time Series. The data has been checked for missing values and it has been found that there are no missing values.

Info of Data:

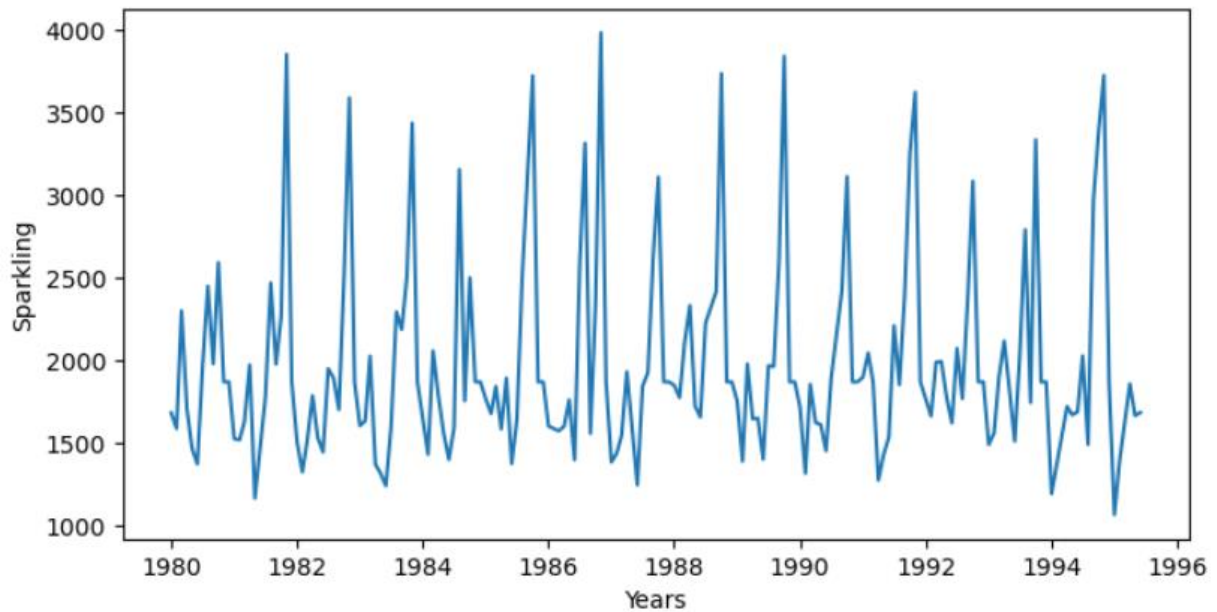
```
<class 'pandas.core.frame.DataFrame'>
DatetimeIndex: 187 entries, 1980-01-01 to 1995-07-01
Data columns (total 1 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Sparkling    187 non-null    int64
dtypes: int64(1)
memory usage: 2.9 KB
```

Description of Data:

[687]:

Sparkling	
count	187.000000
mean	2402.417112
std	1295.111540
min	1070.000000
25%	1605.000000
50%	1874.000000
75%	2549.000000
max	7242.000000

Plotting the data as Time Series:

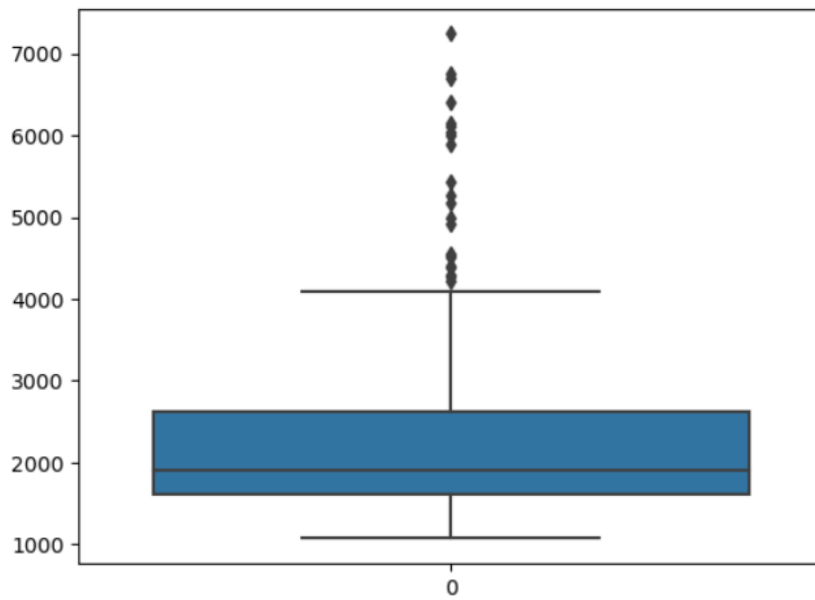


- 2. Perform appropriate Exploratory Data Analysis to understand the data and also perform Decomposition.**

Outliers and Duplicates:

The data was checked for outliers by plotting boxplot of Sparkling data. It has been seen that there are some outliers in the data. The outliers have been imputed with Median value of the data for better Time Series Forecasting.

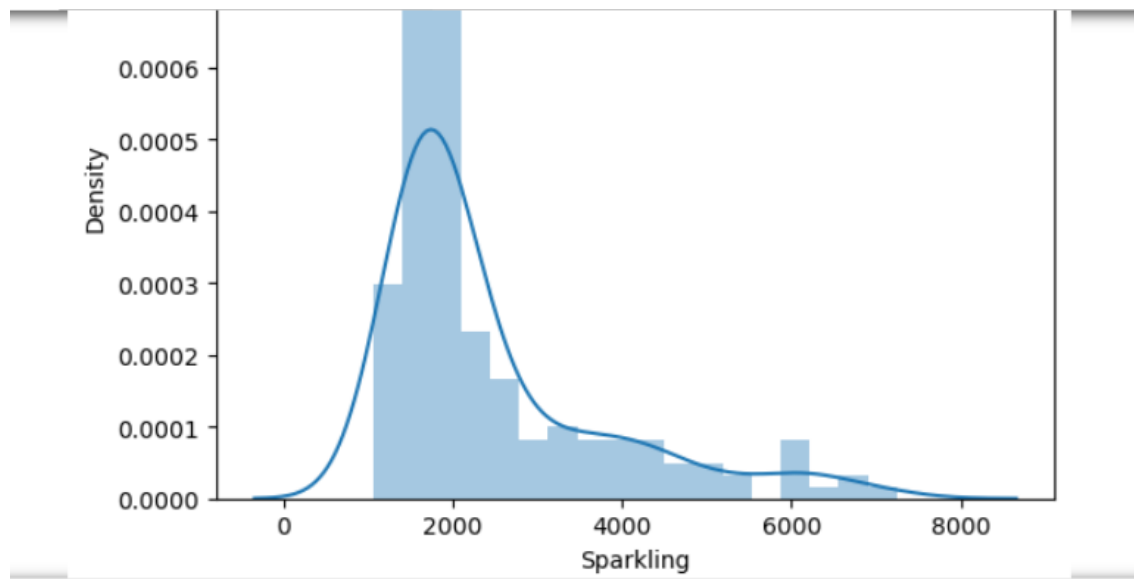
The data has been checked for duplicates and duplicate, if any in the data have been dropped.



Univariate Analysis:

The Histogram of Sparkling Sales have been plotted and it is seen that the data is generally a Normal Distribution with a tail on the right side indicating some outliers.

The Histogram is as under:

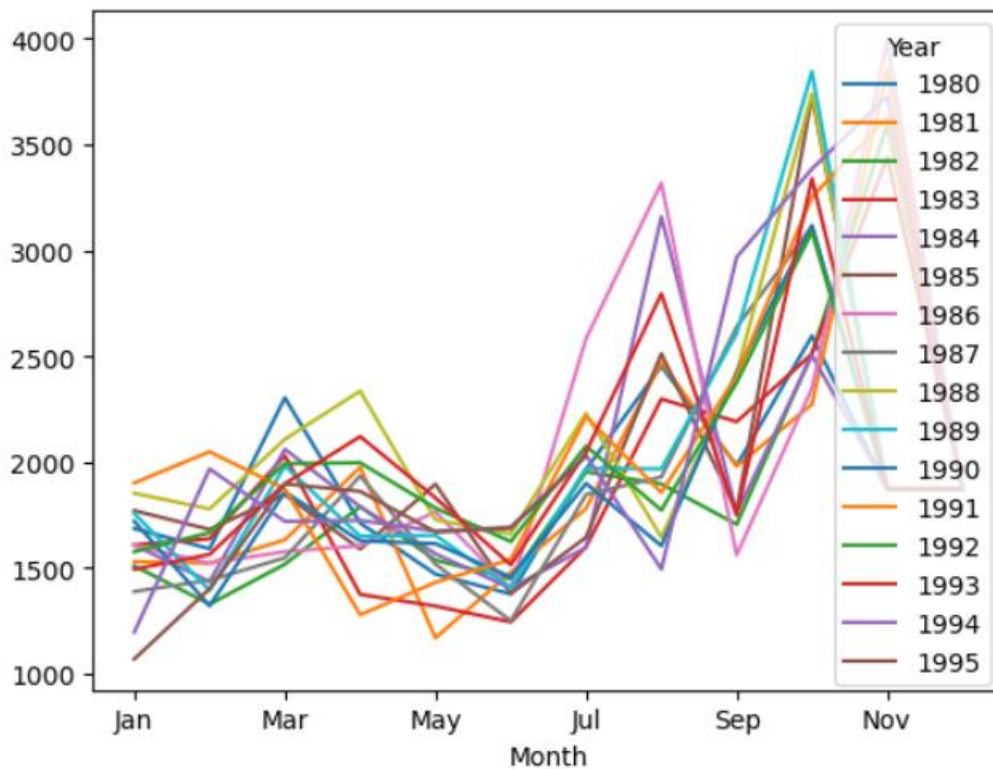


Bivariate Analysis:

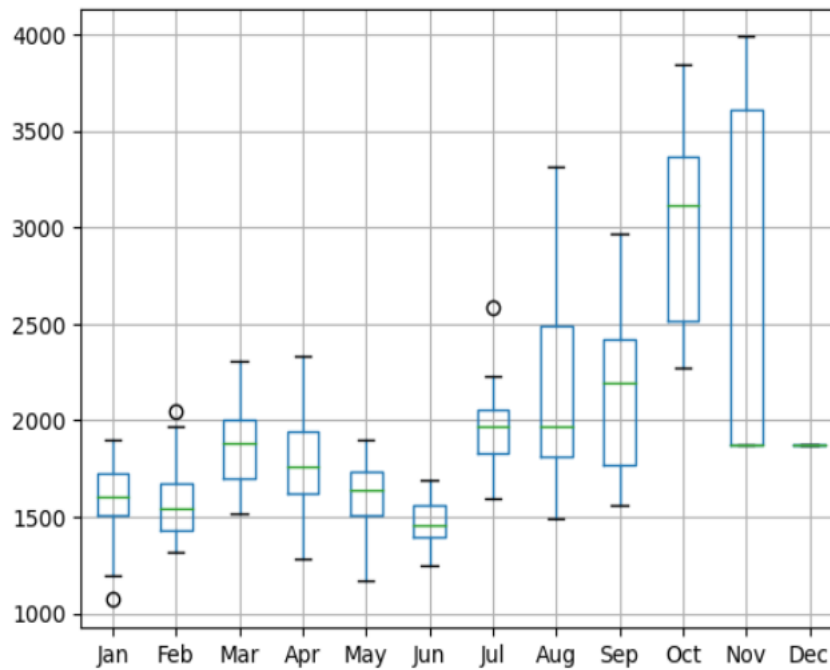
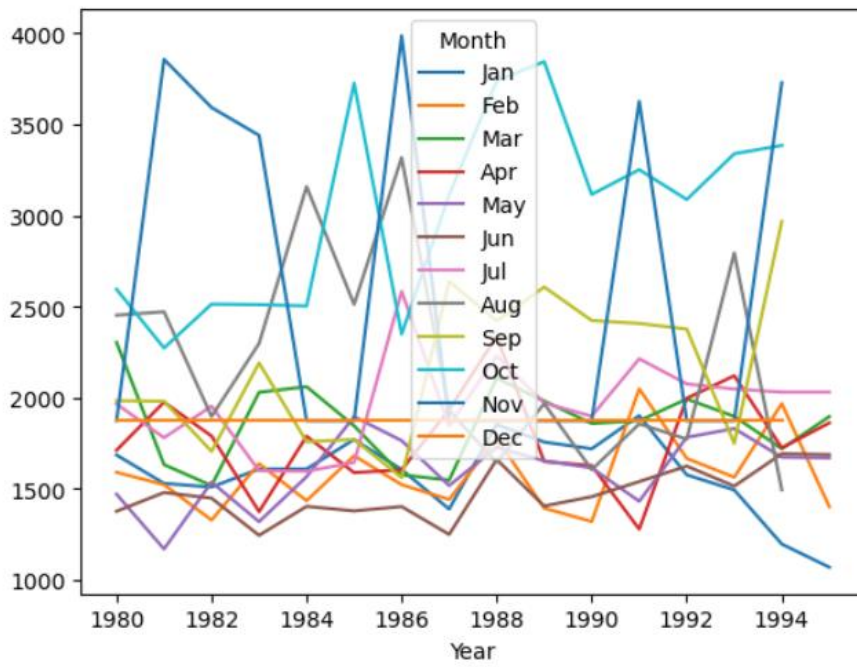
Pivot and Line Charts of Monthly and Yearly Sales of Sparkling Wine have been prepared. Further, boxplot of Yearly Sales has also been prepared and results are as s under:

i) Monthly Sales Data:

Year	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995
Month																
Jan	1686.0	1530.0	1510.0	1609.0	1609.0	1771.0	1606.0	1389.0	1853.0	1757.0	1720.0	1902.0	1577.0	1494.0	1197.0	1070.0
Feb	1591.0	1523.0	1329.0	1638.0	1435.0	1682.0	1523.0	1442.0	1779.0	1394.0	1321.0	2049.0	1667.0	1564.0	1968.0	1402.0
Mar	2304.0	1633.0	1518.0	2030.0	2061.0	1846.0	1577.0	1548.0	2108.0	1982.0	1859.0	1874.0	1993.0	1898.0	1720.0	1897.0
Apr	1712.0	1976.0	1790.0	1375.0	1789.0	1589.0	1605.0	1935.0	2336.0	1650.0	1628.0	1279.0	1997.0	2121.0	1725.0	1862.0
May	1471.0	1170.0	1537.0	1320.0	1567.0	1896.0	1765.0	1518.0	1728.0	1654.0	1615.0	1432.0	1783.0	1831.0	1674.0	1670.0
Jun	1377.0	1480.0	1449.0	1245.0	1404.0	1379.0	1403.0	1250.0	1661.0	1406.0	1457.0	1540.0	1625.0	1515.0	1693.0	1688.0
Jul	1966.0	1781.0	1954.0	1600.0	1597.0	1645.0	2584.0	1847.0	2230.0	1971.0	1899.0	2214.0	2076.0	2048.0	2031.0	2031.0
Aug	2453.0	2472.0	1897.0	2298.0	3159.0	2512.0	3318.0	1930.0	1645.0	1968.0	1605.0	1857.0	1773.0	2795.0	1495.0	NaN
Sep	1984.0	1981.0	1706.0	2191.0	1759.0	1771.0	1562.0	2638.0	2421.0	2608.0	2424.0	2408.0	2377.0	1749.0	2968.0	NaN
Oct	2596.0	2273.0	2514.0	2511.0	2504.0	3727.0	2349.0	3114.0	3740.0	3845.0	3116.0	3252.0	3088.0	3339.0	3385.0	NaN
Nov	1874.0	3857.0	3593.0	3440.0	1874.0	1874.0	3987.0	1874.0	1874.0	1874.0	1874.0	3627.0	1874.0	1874.0	3729.0	NaN
Dec	1874.0	1874.0	1874.0	1874.0	1874.0	1874.0	1874.0	1874.0	1874.0	1874.0	1874.0	1874.0	1874.0	1874.0	1874.0	NaN



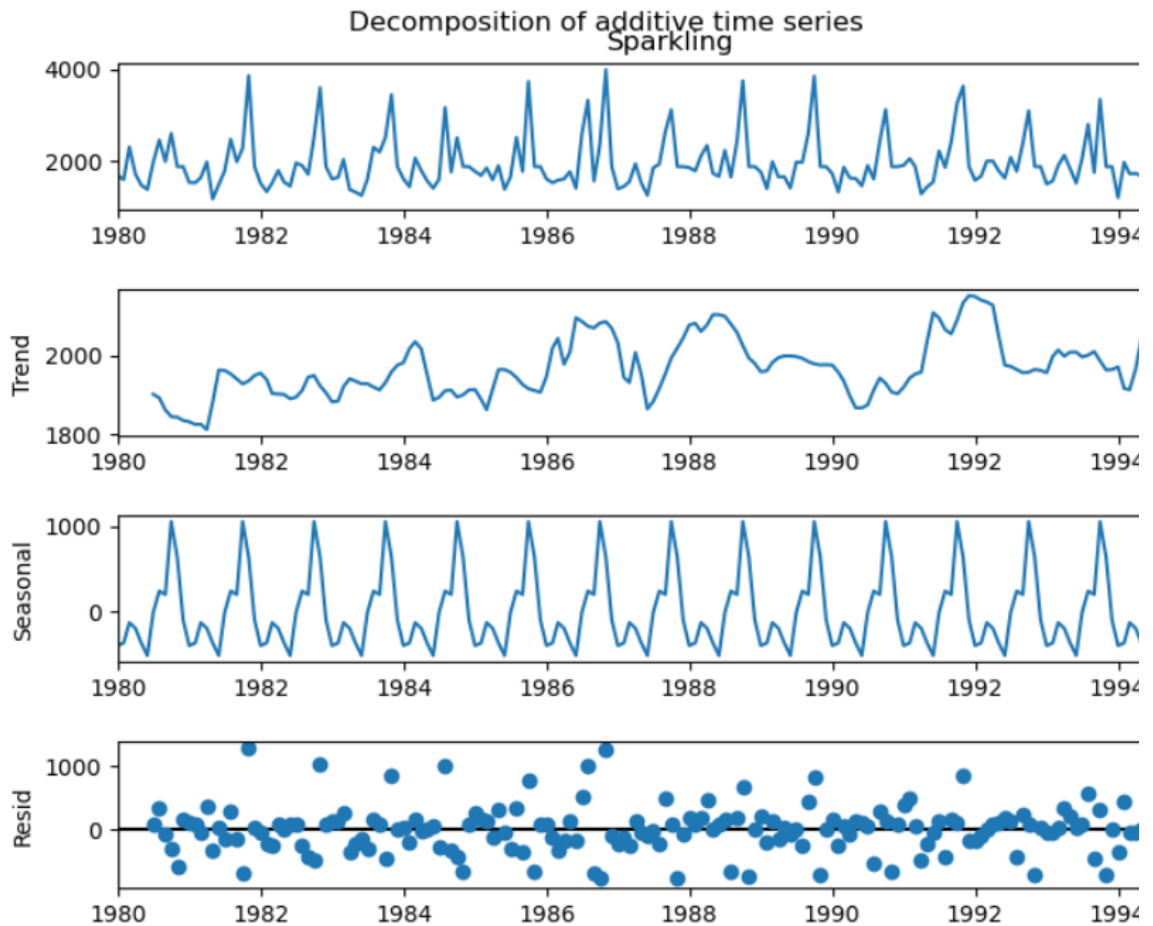
i) Yearly Sales Data:

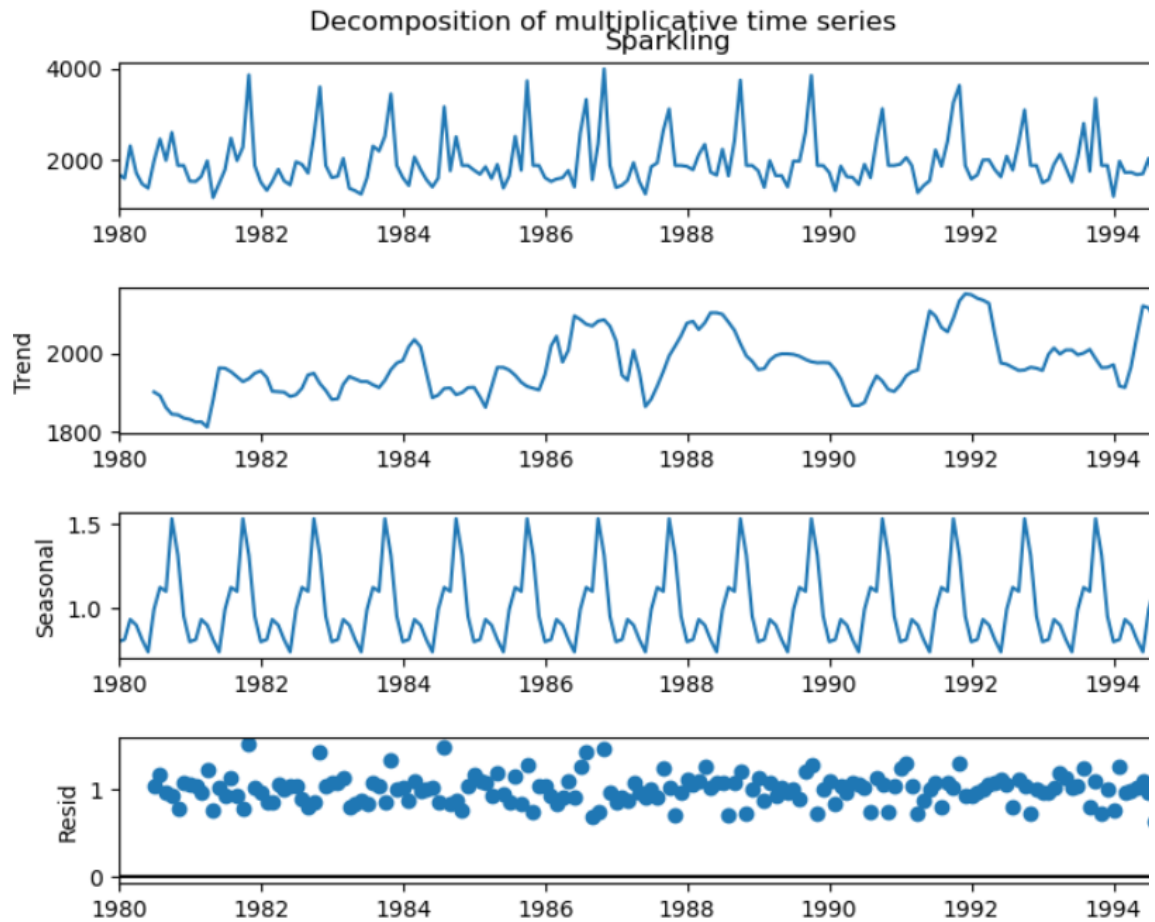


It has been observed that maximum sales are in the months of November and December and minimum sales are in the months May and June. There is prima facie no trend of Year on Year increase in sales.

Decomposition Plots:

The data has been Decomposed for both additive and multiplicative Time Series and the plots are as under:





It may be observed from above that the data has no definite Trend but has Seasonality.

3. Split the Data into training and test. The test data should start in 1991.

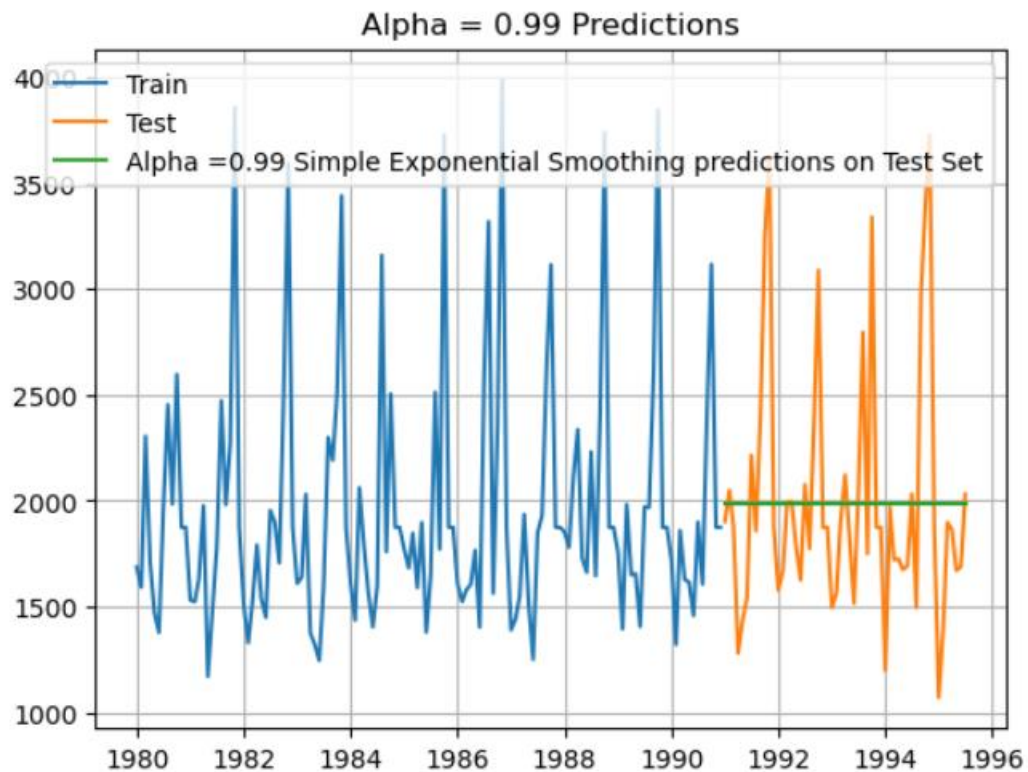
The data has been Split into Training and Test set with Test data starting from 1991. The details are available in Jupyter file appended.

4. Build all the exponential smoothing models on the training data and evaluate the model using RMSE on the test data. Other additional models such as regression, naïve forecast models, simple average models, moving average models should also be built on the training data and check the performance on the test data using RMSE.

Exponential Smoothing Models:

The following Exponential Smoothing Models have been built:

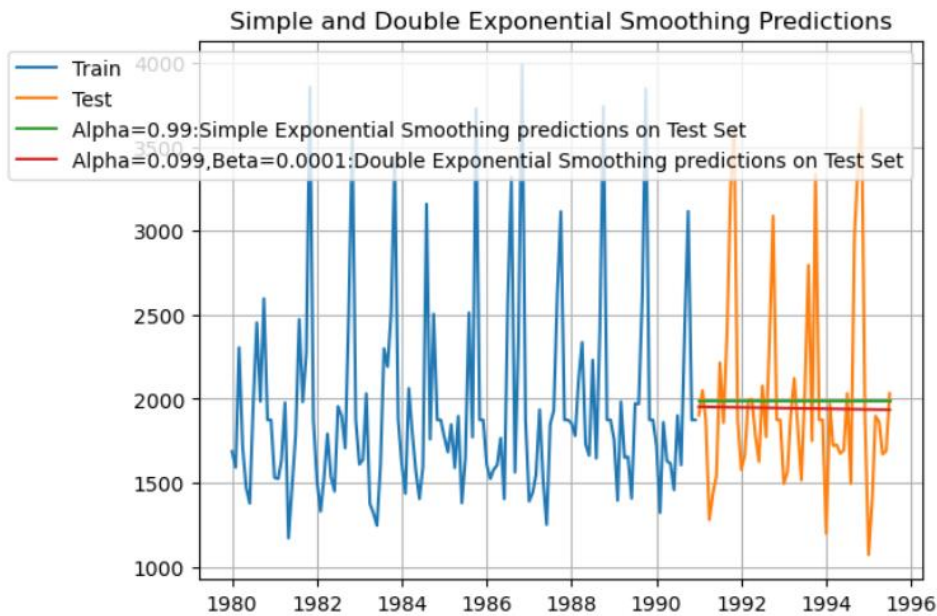
- i) Simple Exponential Smoothing: SES was applied on the data. The predictions on test data and RMSE are as under:



89]:

Test RMSE	
Alpha=0.99,SES	592.073465

- ii) Double Exponential Smoothing: Double Exponential Smoothing (Holt's Method) was applied on the data. The predictions on test data and RMSE are as under:

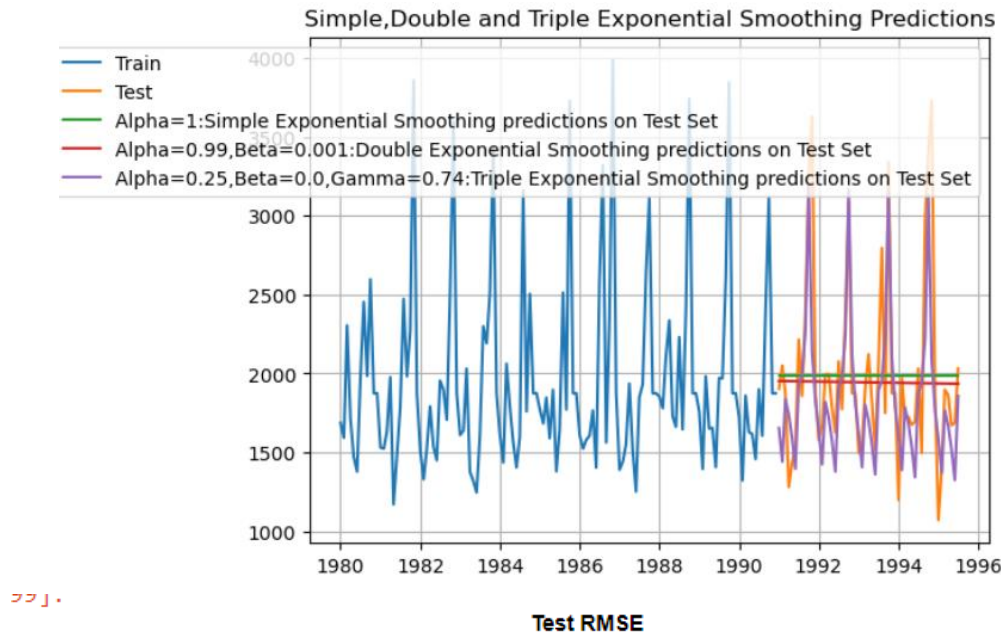


¶]:

Test RMSE

Alpha=0.99,SES	592.073465
Alpha=1,Beta=0.0189:DES	594.973216

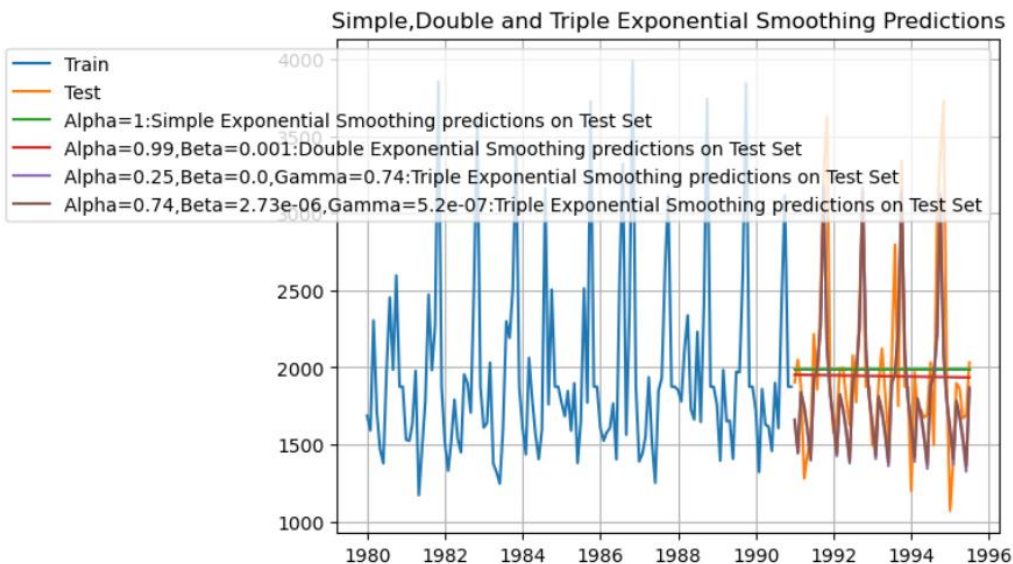
- iii) Triple Exponential Smoothing (with additive errors): Triple Exponential Smoothing (Holt Winter's Method) was applied on the data . The predictions on test data and RMSE are as under:



Alpha=0.99, SES	592.073465
Alpha=1, Beta=0.0189: DES	594.973216
Alpha=0.25, Beta=0.0, Gamma=0.74: TES	417.224017

We can observe that Seasonality component has been added to the predictions.

- iv) Triple Exponential Smoothing (with multiplicative errors): Triple Exponential Smoothing (Holt Winter's Method) was applied on the data. The predictions on test data and RMSE are as under:



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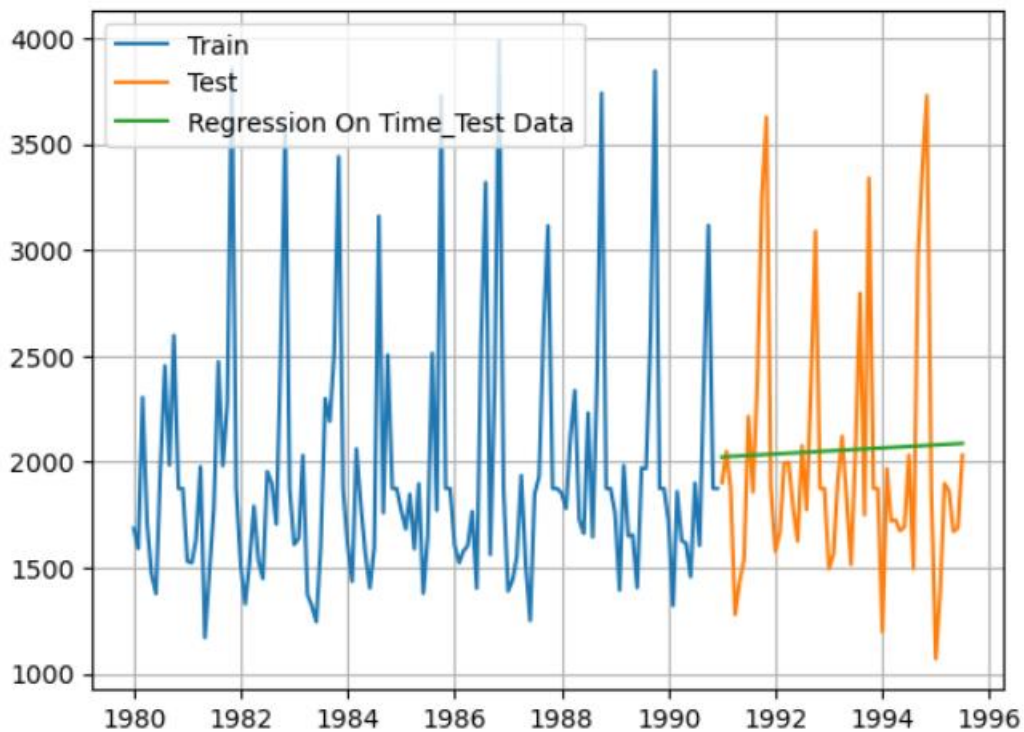
	Test RMSE
Alpha=0.99,SES	592.073465
Alpha=1,Beta=0.0189:DES	594.973216
Alpha=0.25,Beta=0.0,Gamma=0.74:TES	417.224017
Alpha=0.74,Beta=2.73e-06,Gamma=5.2e-07,Gamma=0:TES	415.194198

We can observe that Seasonality component has been added to the predictions.

Other Additional Models:

The following Other Additional Models have been built:

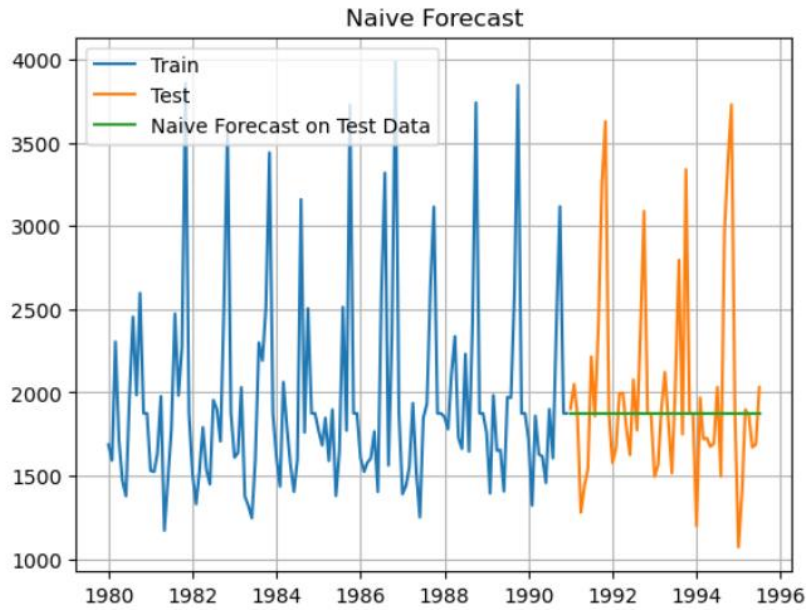
- Linear Regression Model: Linear Regression_was applied on the data. The predictions on test data and RMSE are as under:



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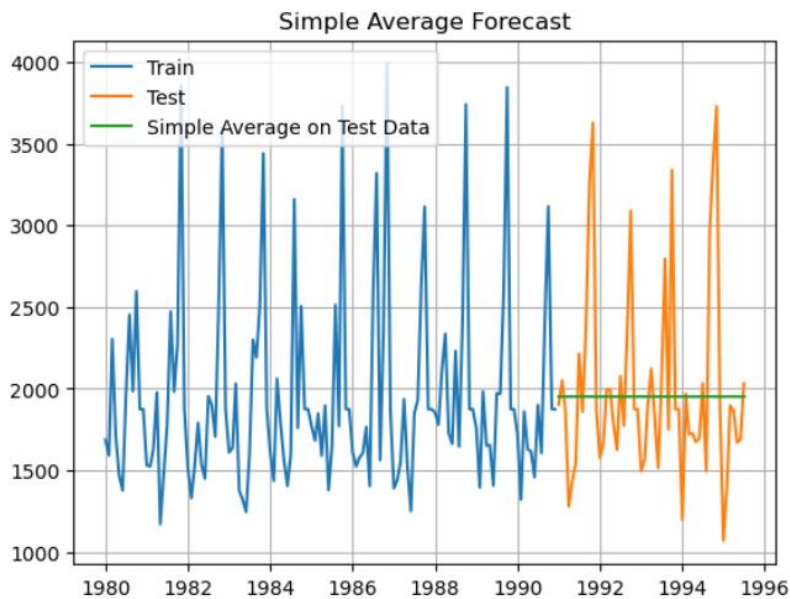
	Test RMSE
RegressionOnTime	594.585092

- ii) Naïve Forecast Model: Naïve Forecast Model was applied on the data The predictions on test data and RMSE are as under:



Test RMSE	
RegressionOnTime	594.585092
NaiveModel	605.921732

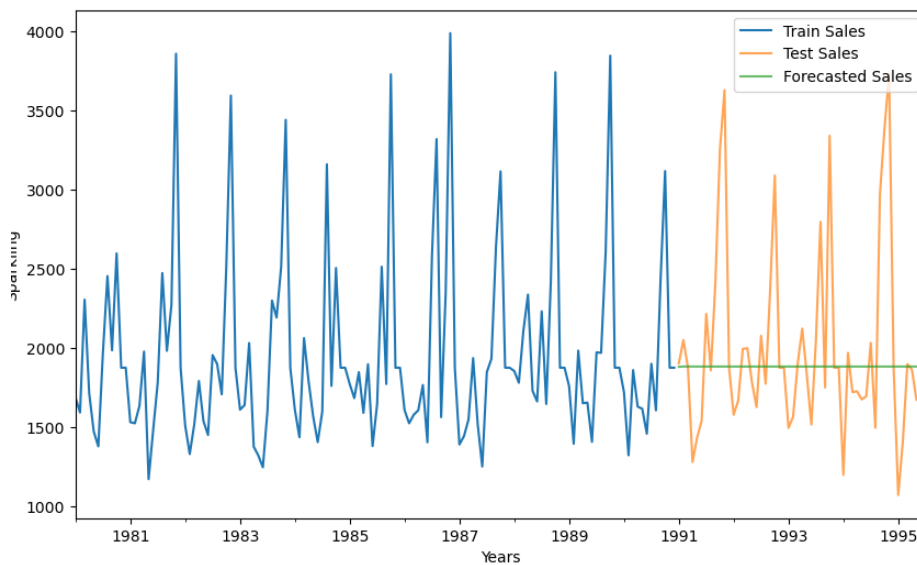
- iii) Simple Average Model: Simple Average Model was applied on the data The predictions on test data and RMSE are as under:



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Test RMSE	
RegressionOnTime	594.585092
NaiveModel	605.921732
SimpleAverageModel	594.215812

- iv) Auto Regressive Model: AR Model was built on the data. The predictions on test data and RMSE are as under:

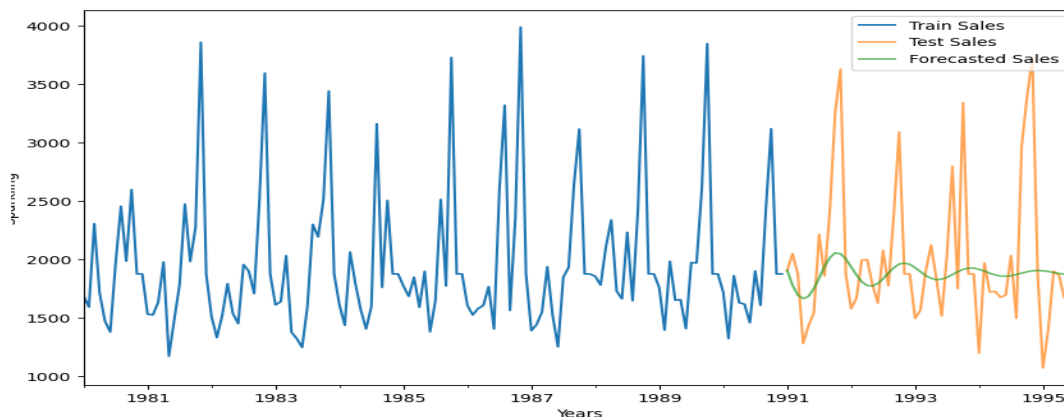


]:

RMSE	
Best AR Model : ARIMA(2,0,0)	604.358653

We can observe a flat output because AR model expresses the current value as a linear combination of past values.

- v) Auto Regressive Moving Average Model: ARMA Model was built on the data. The predictions on test data and RMSE are as under:



	RMSE
Best AR Model : ARIMA(2,0,0)	604.358653
Best ARMA Model : ARIMA(3,0,3)	579.555775

From the output, we can observe that ARMA Model combines both Auto Regressive and Moving Average terms.

5. Check for the stationarity of the data on which the model is being built on using appropriate statistical tests and also mention the hypothesis for the statistical test. If the data is found to be non-stationary, take appropriate steps to make it stationary. Check the new data for stationarity and comment.

Note: Stationarity should be checked at alpha = 0.05.

The stationarity of data has been checked using Dicky Fuller Test.

Null Hypothesis is that the Series is non-stationary and

Alternate Hypothesis is that the series is stationary.

The Test has been done at Alpha = 0.05. The results are as under:

```
# Dickey-Fuller Test - Dicky Fuller Test on the timeseries is run to check for stationarity of data.
# Null Hypothesis H0: Time Series is non-stationary.
# Alternate Hypothesis Ha: Time Series is stationary.
# So Ideally if p-value < 0.05 then null hypothesis: TS is non-stationary is rejected else
# the TS is non-stationary is failed to be rejected

DF test statistic is -3.360
DF test p-value is 0.0124
```

Since p value is less than 0.05, therefore, we reject the Null Hypothesis and Time Series is Stationary.

6. Build an automated version of the ARIMA/SARIMA model in which the parameters are selected using the lowest Akaike Information Criteria (AIC) on the training data and evaluate this model on the test data using RMSE.

Automated versions of ARIMA and SARIMA Models wherein parameters were selected using the lowest Akaike Information Criteria (AIC) were built on the training data and the models were evaluated on the test data and the results are as under:

ARIMA Model: Best Results using lowest AIC of ARIMA Model, Predictions and Root Mean Square Error are as under:

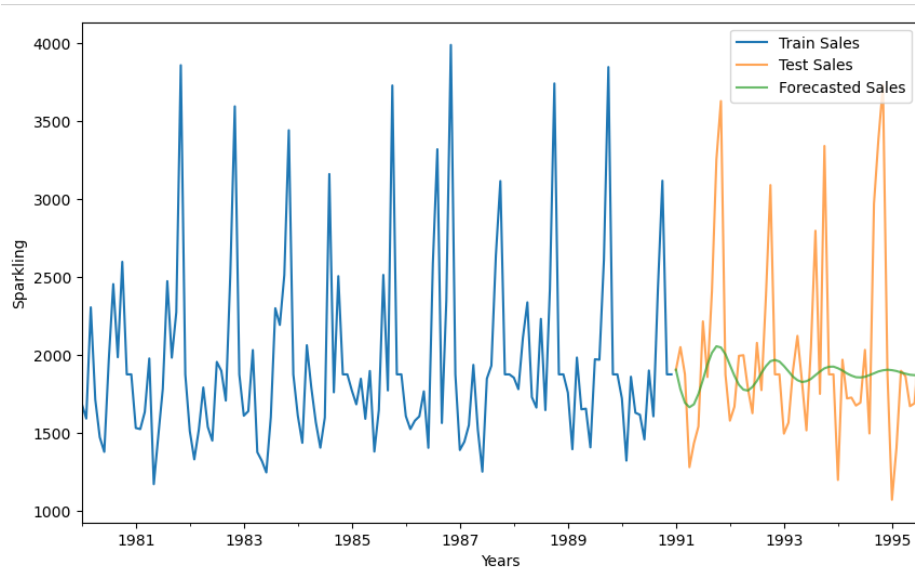
	param	AIC
14	(3, 0, 3)	-232.634490
6	(2, 0, 1)	-218.939533
13	(3, 0, 2)	-217.060140
0	(1, 0, 1)	-207.592434
12	(3, 0, 1)	-207.389271

```
: print(best_results.summary().tables[0])
print(best_results.summary().tables[1])
```

```

=====
SARIMAX Results
=====
Dep. Variable:          Sparkling    No. Observations:          132
Model:                ARIMA(3, 0, 3)  Log Likelihood             124.317
Date:                 Sat, 27 Jan 2024  AIC                        -232.634
Time:                  22:19:54        BIC                        -209.572
Sample:               01-01-1980      HQIC                       -223.263
                   - 12-01-1990
Covariance Type:      opg
=====
=====
              coef    std err          z      P>|z|      [0.025    0.975]
-----
const          3.2748      0.005    690.960      0.000      3.266      3.284
ar.L1           0.6861      0.178      3.845      0.000      0.336      1.036
ar.L2           0.7234      0.259      2.788      0.005      0.215      1.232
ar.L3          -0.8724      0.155     -5.611      0.000     -1.177     -0.568
ma.L1          -0.6529      0.334     -1.957      0.050     -1.307      0.001
ma.L2          -0.8884      0.492     -1.805      0.071     -1.853      0.076
ma.L3           0.7496      0.247      3.031      0.002      0.265      1.234
sigma2          0.0087      0.002      5.219      0.000      0.005      0.012
=====

```



RMSE	
Best AR Model : ARIMA(2,0,0)	604.358653
Best ARMA Model : ARIMA(3,0,3)	579.555775
Best ARIMA Model : ARIMA(3,0,3)	579.555775

SARIMA Model: Best Results using lowest AIC of SARIMA Model, Predictions and Root Mean Square Error are as under:

	param	seasonal	AIC
0	(1, 0, 1)	(1, 0, 1, 12)	-249.512697
18	(1, 0, 2)	(1, 0, 1, 12)	-246.355744
216	(3, 0, 1)	(1, 0, 1, 12)	-245.256571
54	(1, 1, 1)	(1, 0, 1, 12)	-241.462741
36	(1, 0, 3)	(1, 0, 1, 12)	-241.131061

```

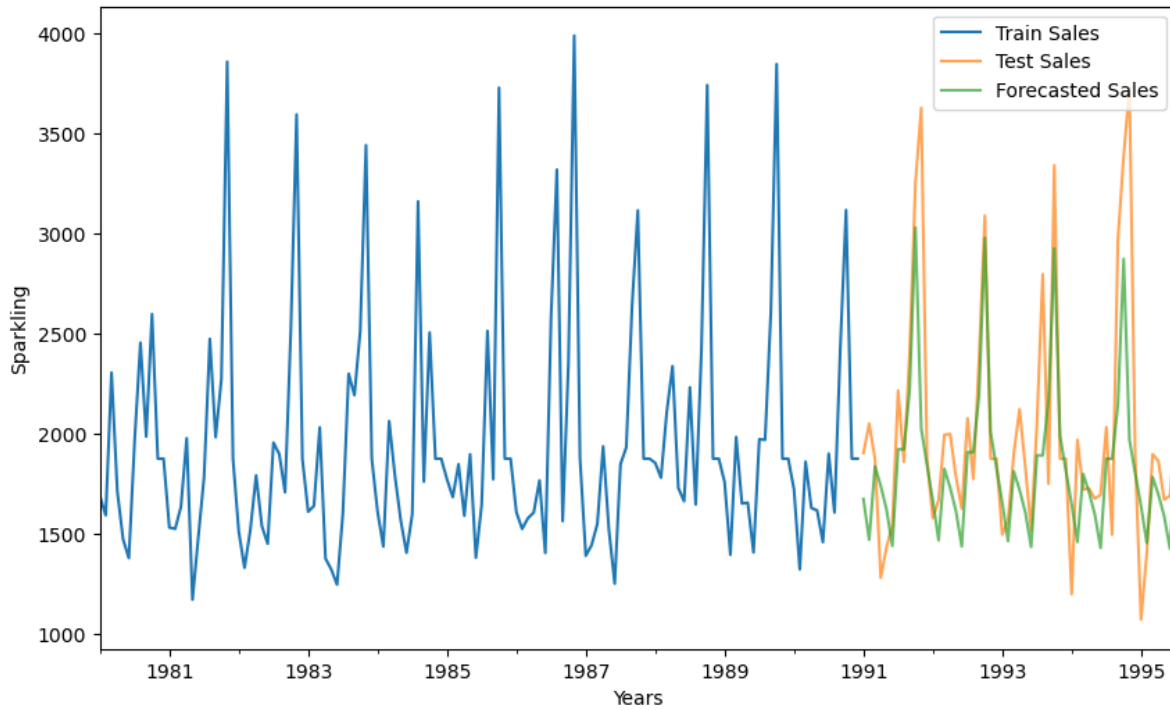
=====
SARIMAX Results
=====
Dep. Variable:          Sparkling    No. Observations:      132
Model:                SARIMAX(1, 0, 1)x(1, 0, 1, 12)    Log Likelihood        134.283
Date:                  Sat, 27 Jan 2024    AIC                   -258.567
Time:                  22:34:17    BIC                   -244.153
Sample:                01-01-1980    HQIC                  -252.710
                    - 12-01-1990
Covariance Type:      opg
=====
=====
              coef      std err          z      P>|z|      [0.025      0.975]
-----
ar.L1          0.9996         0.001    1059.626      0.000         0.998         1.001
ma.L1         -0.9487         0.068    -13.967      0.000        -1.082        -0.816
ar.S.L12       0.9791         0.017     59.117      0.000         0.947         1.012
ma.S.L12      -0.7138         0.083     -8.596      0.000        -0.877        -0.551
sigma2         0.0063         0.001      7.509      0.000         0.005         0.008
=====

```

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	Month	Year	Sparkling	date	Sparkling_sales_forecasted
1991-01-01	Jan	1991	1902	1991-01-01	1672.235721
1991-02-01	Feb	1991	2049	1991-02-01	1468.832620
1991-03-01	Mar	1991	1874	1991-03-01	1835.474946
1991-04-01	Apr	1991	1279	1991-04-01	1738.178581
1991-05-01	May	1991	1432	1991-05-01	1618.976609
1991-06-01	Jun	1991	1540	1991-06-01	1438.157210
1991-07-01	Jul	1991	2214	1991-07-01	1921.384929
1991-08-01	Aug	1991	1857	1991-08-01	1920.469408
1991-09-01	Sep	1991	2408	1991-09-01	2217.045485
1991-10-01	Oct	1991	3252	1991-10-01	3029.021716
1991-11-01	Nov	1991	3627	1991-11-01	2024.588999
1991-12-01	Dec	1991	1874	1991-12-01	1841.092605
1992-01-01	Jan	1992	1577	1992-01-01	1664.850181
1992-02-01	Feb	1992	1667	1992-02-01	1466.226830
1992-03-01	Mar	1992	1993	1992-03-01	1823.601685
1992-04-01	Apr	1992	1997	1992-04-01	1728.799300
1992-05-01	May	1992	1783	1992-05-01	1612.537572
1992-06-01	Jun	1992	1625	1992-06-01	1435.901611
1992-07-01	Jul	1992	2076	1992-07-01	1906.684727
1992-08-01	Aug	1992	1773	1992-08-01	1905.683415
1992-09-01	Sep	1992	2377	1992-09-01	2193.257141
1992-10-01	Oct	1992	3088	1992-10-01	2976.874269
1992-11-01	Nov	1992	1874	1992-11-01	2006.433847
1992-12-01	Dec	1992	1874	1992-12-01	1828.101007
1993-01-01	Jan	1993	1494	1993-01-01	1656.484068
1993-02-01	Feb	1993	1564	1993-02-01	1462.650382
1993-03-01	Mar	1993	1898	1993-03-01	1810.776764
1993-04-01	Apr	1993	2121	1993-04-01	1718.456307
1993-05-01	May	1993	1831	1993-05-01	1605.129323

1993-06-01	Jun	1993	1515	1993-06-01	1432.689650
1993-07-01	Jul	1993	2048	1993-07-01	1891.072061
1993-08-01	Aug	1993	2795	1993-08-01	1889.989356
1993-09-01	Sep	1993	1749	1993-09-01	2168.690693
1993-10-01	Oct	1993	3339	1993-10-01	2924.633938
1993-11-01	Nov	1993	1874	1993-11-01	1987.421673
1993-12-01	Dec	1993	1874	1993-12-01	1814.197595
1994-01-01	Jan	1994	1197	1994-01-01	1647.178973
1994-02-01	Feb	1994	1968	1994-02-01	1458.135504
1994-03-01	Mar	1994	1720	1994-03-01	1797.048611
1994-04-01	Apr	1994	1725	1994-04-01	1707.194192
1994-05-01	May	1994	1674	1994-05-01	1596.791387
1994-06-01	Jun	1994	1693	1994-06-01	1428.552442
1994-07-01	Jul	1994	2031	1994-07-01	1874.598618
1994-08-01	Aug	1994	1495	1994-08-01	1873.438906
1994-09-01	Sep	1994	2968	1994-09-01	2143.406486
1994-10-01	Oct	1994	3385	1994-10-01	2872.367160
1994-11-01	Nov	1994	3729	1994-11-01	1967.607539
1994-12-01	Dec	1994	1874	1994-12-01	1799.431297
1995-01-01	Jan	1995	1070	1995-01-01	1636.976937
1995-02-01	Feb	1995	1402	1995-02-01	1452.715409
1995-03-01	Mar	1995	1897	1995-03-01	1782.465597
1995-04-01	Apr	1995	1862	1995-04-01	1695.057756
1995-05-01	May	1995	1670	1995-05-01	1587.563825
1995-06-01	Jun	1995	1688	1995-06-01	1423.522106
1995-07-01	Jul	1995	2031	1995-07-01	1857.315714



RMSE	
Best AR Model : ARIMA(2,0,0)	604.358653
Best ARMA Model : ARIMA(3,0,3)	579.555775
Best ARIMA Model : ARIMA(3,0,3)	579.555775
Best SARIMA Model : SARIMAX(1, 0, 1)x(1, 0, 1, 12)	435.356533

It is observed that Best SARIMA Model (1,0,1)x(1,0,1,12) with RMSE of 435.36 is the Best Model.

7. Build a table with all the models built along with their corresponding parameters and the respective RMSE values on the test data.

Comparison of Models: Comparison of all the aforesaid Models is tabulated as under:

S.No.	Model	Root Mean Square Error (RMSE) on Test Data
1.	Simple Exponential Smoothing Model	592.07
2.	Double Exponential Smoothing Model	594.97
3.	Triple Exponential Smoothing (with additive errors)	417.22
4.	Triple Exponential Smoothing (with multiplicative errors)	415.19
5.	Linear Regression	594.58
6.	Naïve Model	605.92
7.	Simple Average Model	604.36
8.	Auto Regressive (AR) Model	1330.37
9.	Auto Regressive Moving Average (ARMA) Model	579.55
10.	ARIMA Model	579.55
11.	SARIMA Model	435.36

It is seen from above that SARIMA Model with Lowest RMSE is the best Model which captures both trend and seasonality in an optimal manner.

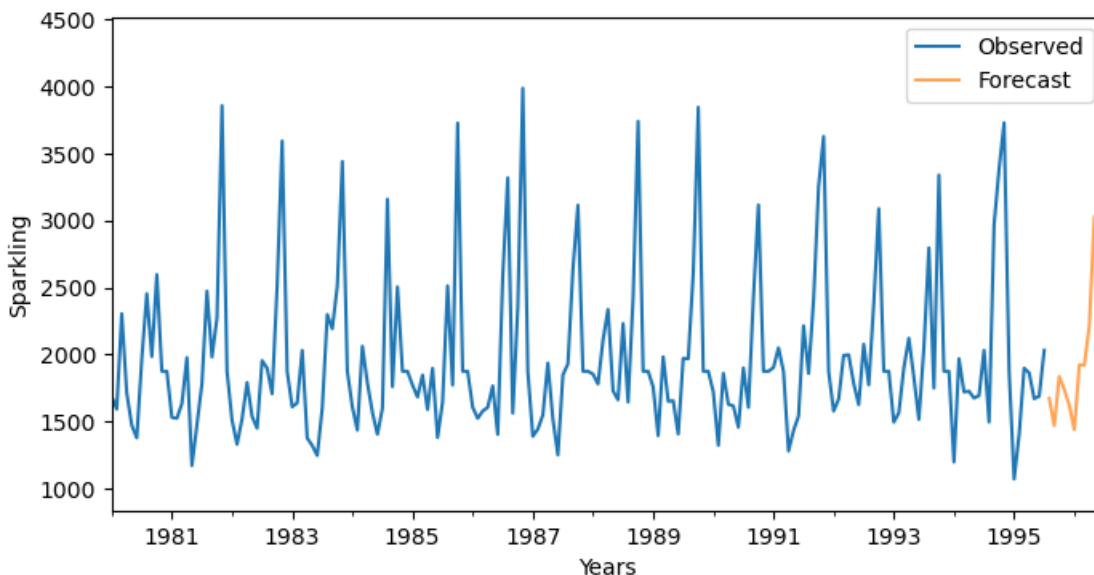
8. Based on the model-building exercise, build the most optimum model(s) on the complete data and predict 12 months into the future with appropriate confidence intervals/bands.

Forecasting was done for the 12 months into future using best SARIMA Model with lowest AIC and the results and plot of predictions with 95% Confidence Interval and 99% Confidence Interval is as under:

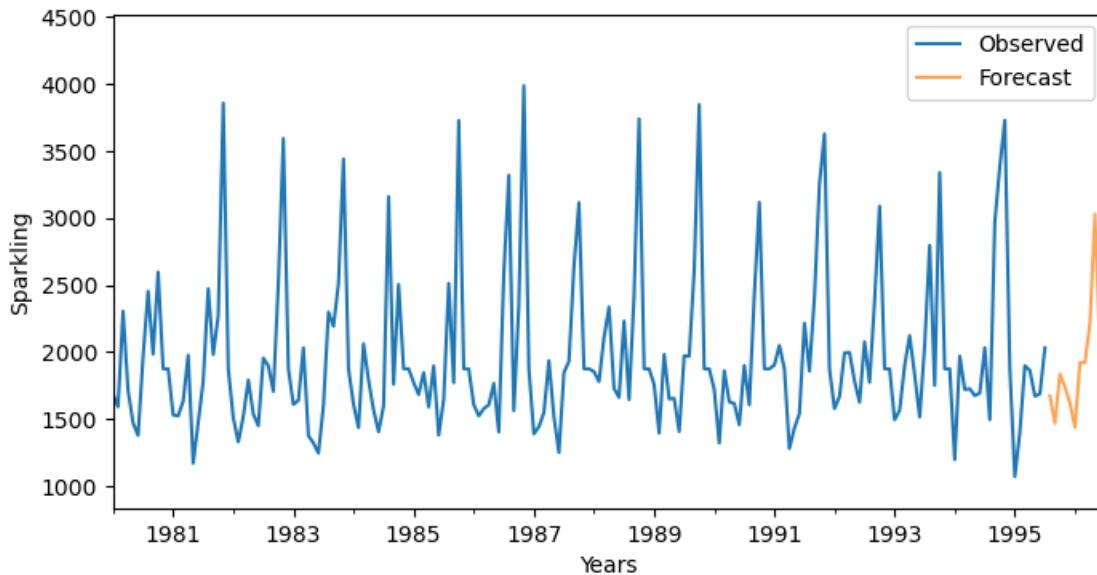
['J']

	forecast	lower_ci_95	upper_ci_95	lower_ci_99	upper_ci_99
1995-08-01	1672.235721	1168.875685	2392.360749	1168.875685	2392.360749
1995-09-01	1468.832620	1026.222701	2102.340227	1026.222701	2102.340227
1995-10-01	1835.474946	1281.789376	2628.332191	1281.789376	2628.332191
1995-11-01	1738.178581	1213.282449	2490.157822	1213.282449	2490.157822
1995-12-01	1618.976609	1129.556149	2320.455927	1129.556149	2320.455927
1996-01-01	1438.157210	1002.937232	2062.238886	1002.937232	2062.238886
1996-02-01	1921.384929	1339.313737	2756.426626	1339.313737	2756.426626
1996-03-01	1920.469408	1338.062189	2756.376181	1338.062189	2756.376181
1996-04-01	2217.045485	1543.991402	3183.496151	1543.991402	3183.496151
1996-05-01	3029.021716	2108.504033	4351.413329	2108.504033	4351.413329
1996-06-01	2024.588999	1408.675954	2909.796680	1408.675954	2909.796680
1996-07-01	1841.092605	1280.419985	2647.273566	1280.419985	2647.273566

Plot of predictions with 95% Confidence Interval is as under:



Plot of predictions with 99% Confidence Interval is as under:



9. Comment on the model thus built and report your findings and suggest the measures that the company should be taking for future sales.

It is seen that SARIMA Model with the lowest AIC of -249.51 and RMSE of 435.36 is the best Model which can predict both trend and seasonality of the data while also taking care of the stationarity factor.

Based on the Model selected, the suggested measures for the Company to be taken for future 12 months sales are as under:

- i) It has been seen from predictions that sales of Sparkling Wine start increasing in the Month of February 1996 and continue till July 1996 with peak in the Month of May 1996. It is required for the Company to keep high inventory of raw materials for production from January/February onwards.
- ii) Adequate planning and deployment of resources is required to meet the demand during peak sales period.
- iii) Attractive discounts and promotional activities to be undertaken to boost wine sales in lean months.
- iv) Lowest sales are in the month of January 1996. Hence, the routine maintenance of machinery should be planned in January 1996.
- v) Export activities may be undertaken during lean period to boost sales.

